

Titration report



Method

Method ID	CDC_GranTED	Release state	Released
Name	Titration with HCl - Gran Verification	Modified by	PredefinedUser
Modified on	2/3/2026 3:42:13 PM		

Summary

Samples

Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.005523 mL
		Concentration R2	0.10011 mol/L
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.012149 mL
		Concentration R2	0.10024 mol/L
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.007243 mL
		Concentration R2	0.10014 mol/L

Statistics

R1

Mean	5.008305 mL	Minimum	5.005523 mL
Standard deviation	0.003438 mL	Rel. standard deviation	0.069 %
Total samples	3	Samples excluded	0

R2

Mean	0.10016 mol/L	Minimum	0.10011 mol/L
Standard deviation	0.00007 mol/L	Rel. standard deviation	0.068 %
Total samples	3	Samples excluded	0

Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.005523 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10011 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.012149 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10024 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.007243 mL	OK

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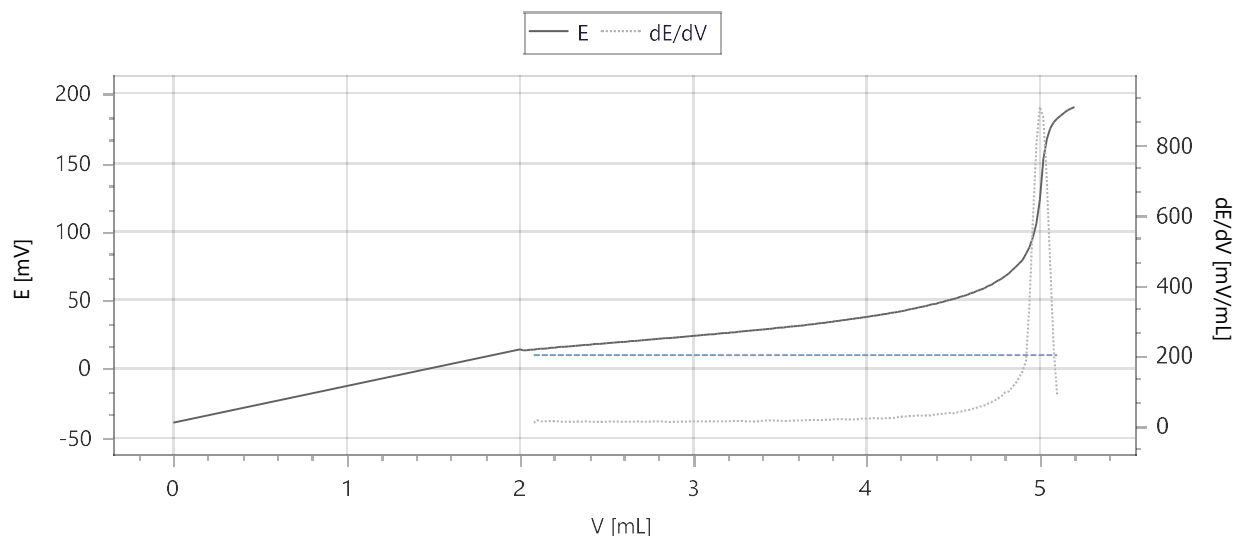
0.1 M TRIS - 5 mL
in 60 mL H₂O

Concentration R2

0.10014 mol/L OK

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3



Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-39.2		25.0			
12	2.0000	14.2	2.0000	25.0	53.4		
45	2.0200	13.5	0.0200	25.0	-0.7		
48	2.0400	13.6	0.0200	25.0	0.1		
52	2.0600	13.9	0.0200	25.0	0.3		
55	2.0800	14.1	0.0200	25.0	0.2	7.42	
58	2.1000	14.4	0.0200	25.0	0.3	14.42	
61	2.1200	14.6	0.0200	25.0	0.2	11.50	
64	2.1400	14.8	0.0200	25.0	0.2	11.21	
67	2.1600	15.0	0.0200	25.0	0.2	12.08	
70	2.1800	15.3	0.0200	25.0	0.3	11.71	
73	2.2000	15.5	0.0200	25.0	0.2	11.77	
76	2.2200	15.8	0.0200	25.0	0.3	11.31	
79	2.2400	16.0	0.0200	25.0	0.2	10.71	
82	2.2600	16.1	0.0200	25.0	0.1	10.22	
85	2.2800	16.4	0.0200	25.0	0.3	9.99	
88	2.3000	16.6	0.0200	25.0	0.2	10.48	
91	2.3200	16.8	0.0200	25.0	0.2	10.74	
94	2.3400	17.0	0.0200	25.0	0.2	10.77	
97	2.3600	17.2	0.0200	25.0	0.2	10.51	
100	2.3800	17.5	0.0200	25.0	0.3	11.29	
104	2.4000	17.6	0.0200	25.0	0.1	10.71	
106	2.4200	17.9	0.0200	25.0	0.3	10.28	
110	2.4400	18.1	0.0200	25.0	0.2	9.99	
113	2.4600	18.3	0.0200	25.0	0.2	9.94	

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116	2.4800	18.4	0.0200	25.0	0.1	9.99
119	2.5000	18.7	0.0200	25.0	0.3	10.28
122	2.5200	18.9	0.0200	25.0	0.2	11.01
125	2.5400	19.1	0.0200	25.0	0.2	10.71
128	2.5600	19.3	0.0200	25.0	0.2	10.86
131	2.5800	19.6	0.0200	25.0	0.3	9.93
134	2.6000	19.7	0.0200	25.0	0.1	10.20
137	2.6200	19.9	0.0200	25.0	0.2	10.31
140	2.6400	20.2	0.0200	25.0	0.3	10.48
143	2.6600	20.3	0.0200	25.0	0.1	10.71
146	2.6800	20.6	0.0200	25.0	0.3	11.09
149	2.7000	20.8	0.0200	25.0	0.2	10.19
152	2.7200	21.0	0.0200	25.0	0.2	10.22
155	2.7400	21.2	0.0200	25.0	0.2	10.80
158	2.7600	21.4	0.0200	25.0	0.2	10.71
161	2.7800	21.6	0.0200	25.0	0.2	11.29
164	2.8000	21.9	0.0200	25.0	0.3	11.29
168	2.8200	22.1	0.0200	25.0	0.2	11.00
171	2.8400	22.3	0.0200	25.0	0.2	10.51
174	2.8600	22.5	0.0200	25.0	0.2	9.70
177	2.8800	22.7	0.0200	25.0	0.2	9.99
180	2.9000	22.9	0.0200	25.0	0.2	10.23
183	2.9200	23.1	0.0200	25.0	0.2	10.48
186	2.9400	23.3	0.0200	25.0	0.2	10.80
189	2.9600	23.6	0.0200	25.0	0.3	10.71
192	2.9800	23.7	0.0200	25.0	0.1	10.99
195	3.0000	24.0	0.0200	25.0	0.3	11.31
198	3.0200	24.2	0.0200	25.0	0.2	11.48
201	3.0400	24.4	0.0200	25.0	0.2	11.22
204	3.0600	24.7	0.0200	25.0	0.3	11.57
208	3.0800	24.9	0.0200	25.0	0.2	10.93
211	3.1000	25.1	0.0200	25.0	0.2	11.80
214	3.1200	25.3	0.0200	25.0	0.2	11.51
217	3.1400	25.6	0.0200	25.0	0.3	11.48
220	3.1600	25.8	0.0200	25.0	0.2	11.51
223	3.1800	26.1	0.0200	25.0	0.3	11.21
226	3.2000	26.2	0.0200	25.0	0.1	11.50
229	3.2200	26.5	0.0200	25.0	0.3	12.02
232	3.2400	26.7	0.0200	25.0	0.2	12.77
236	3.2600	27.0	0.0200	25.0	0.3	12.51
239	3.2800	27.3	0.0200	25.0	0.3	12.57
242	3.3000	27.5	0.0200	25.0	0.2	11.73
245	3.3200	27.7	0.0200	25.0	0.2	11.21
248	3.3400	27.9	0.0200	25.0	0.2	10.98
251	3.3600	28.2	0.0200	25.0	0.3	10.92
254	3.3800	28.4	0.0200	25.0	0.2	11.70
257	3.4000	28.6	0.0200	25.0	0.2	12.81
260	3.4200	28.9	0.0200	25.0	0.3	13.79
263	3.4400	29.1	0.0200	25.0	0.2	13.99
266	3.4600	29.5	0.0200	25.0	0.4	14.07
269	3.4800	29.8	0.0200	25.0	0.3	13.78
272	3.5000	30.0	0.0200	25.0	0.2	12.95

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276	3.5200	30.2	0.0200	25.0	0.2	12.49
278	3.5400	30.5	0.0200	25.0	0.3	12.49
282	3.5600	30.8	0.0200	25.0	0.3	12.92
285	3.5800	31.0	0.0200	25.0	0.2	13.49
288	3.6000	31.3	0.0200	25.0	0.3	14.09
291	3.6200	31.6	0.0200	25.0	0.3	13.92
294	3.6400	31.8	0.0200	25.0	0.2	14.49
297	3.6600	32.2	0.0200	25.0	0.4	15.09
300	3.6800	32.5	0.0200	25.0	0.3	15.20
303	3.7000	32.7	0.0200	25.0	0.2	15.52
306	3.7200	33.1	0.0200	25.0	0.4	15.28
309	3.7400	33.4	0.0200	25.0	0.3	15.19
312	3.7600	33.7	0.0200	25.0	0.3	15.74
315	3.7800	34.0	0.0200	25.0	0.3	15.99
318	3.8000	34.3	0.0200	25.0	0.3	16.50
321	3.8200	34.7	0.0200	25.0	0.4	18.09
324	3.8400	35.0	0.0200	25.0	0.3	17.71
327	3.8600	35.4	0.0200	25.0	0.4	17.56
330	3.8800	35.8	0.0200	25.0	0.4	17.01
334	3.9000	36.1	0.0200	25.0	0.3	16.93
336	3.9200	36.3	0.0200	25.0	0.2	16.72
340	3.9400	36.8	0.0200	25.0	0.5	16.97
343	3.9600	37.1	0.0200	25.0	0.3	18.50
346	3.9800	37.5	0.0200	25.0	0.4	19.24
349	4.0000	37.8	0.0200	25.0	0.3	19.55
352	4.0200	38.3	0.0200	25.0	0.5	19.80
355	4.0400	38.7	0.0200	25.0	0.4	20.21
358	4.0600	39.0	0.0200	25.0	0.3	20.00
361	4.0800	39.5	0.0200	25.0	0.5	19.79
364	4.1000	39.9	0.0200	25.0	0.4	19.90
367	4.1200	40.2	0.0200	25.0	0.3	20.74
370	4.1400	40.7	0.0200	25.0	0.5	20.98
373	4.1600	41.1	0.0200	25.0	0.4	21.69
376	4.1800	41.6	0.0200	25.0	0.5	23.53
380	4.2000	42.0	0.0200	25.0	0.4	24.49
382	4.2200	42.5	0.0200	25.0	0.5	25.29
386	4.2400	43.1	0.0200	25.0	0.6	26.29
389	4.2600	43.6	0.0200	25.0	0.5	26.71
392	4.2800	44.1	0.0200	25.0	0.5	27.30
395	4.3000	44.7	0.0200	25.0	0.6	28.08
398	4.3200	45.2	0.0200	25.0	0.5	27.64
402	4.3400	45.8	0.0200	25.0	0.6	28.26
405	4.3600	46.4	0.0200	25.0	0.6	28.22
408	4.3800	47.0	0.0200	25.0	0.6	29.42
411	4.4000	47.4	0.0200	25.0	0.4	31.31
414	4.4200	48.2	0.0200	25.0	0.8	31.99
417	4.4400	48.8	0.0200	25.0	0.6	32.96
421	4.4600	49.6	0.0200	25.0	0.8	34.03
426	4.4800	50.2	0.0200	25.0	0.6	35.58
429	4.5000	50.8	0.0200	25.0	0.6	34.29
432	4.5200	51.6	0.0200	25.0	0.8	36.52
435	4.5400	52.4	0.0200	25.0	0.8	38.45

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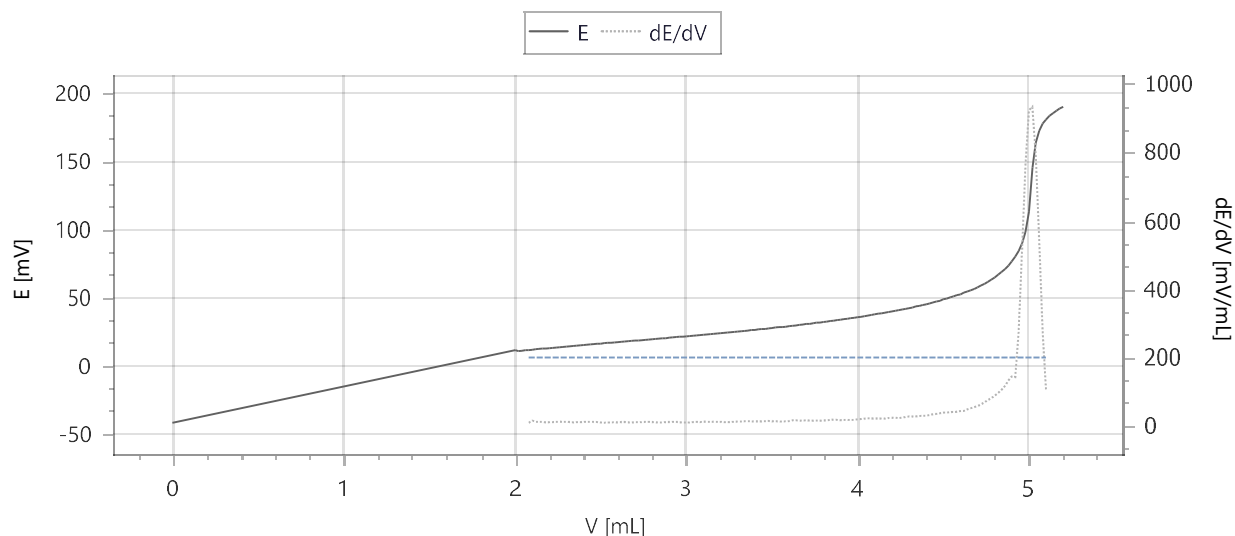
438	4.5600	53.2	0.0200	25.0	0.8	41.13	
441	4.5800	53.8	0.0200	25.0	0.6	43.87	
444	4.6000	55.0	0.0200	25.0	1.2	44.34	
447	4.6200	55.8	0.0200	25.0	0.8	49.01	
450	4.6400	56.8	0.0200	25.0	1.0	50.95	
453	4.6600	57.9	0.0200	25.0	1.1	53.90	
456	4.6800	58.8	0.0200	25.0	0.9	56.96	
461	4.7000	60.4	0.0200	25.0	1.6	61.63	
464	4.7200	61.2	0.0200	25.0	0.8	65.79	
469	4.7400	62.9	0.0200	25.0	1.7	71.27	
474	4.7600	64.3	0.0200	25.0	1.4	77.26	
476	4.7800	65.8	0.0200	25.0	1.5	83.72	
481	4.8000	67.7	0.0200	25.0	1.9	94.14	
485	4.8200	69.6	0.0200	25.0	1.9	96.52	
488	4.8400	71.7	0.0200	25.0	2.1	108.90	
494	4.8600	74.1	0.0200	25.0	2.4	119.32	
497	4.8800	76.7	0.0200	25.0	2.6	135.37	
501	4.9000	79.3	0.0200	25.0	2.6	154.72	
506	4.9200	84.0	0.0200	25.0	4.7	183.44	
509	4.9400	88.3	0.0200	25.0	4.3	346.74	
516	4.9600	95.5	0.0200	25.0	7.2	586.08	
524	4.9800	105.5	0.0200	25.0	10.0	799.53	
537	5.0000	123.0	0.0200	25.0	17.5	907.87	
	5.005523	131.3				908.66	EQP
545	5.0200	153.0	0.0200	25.0	30.0	876.31	
550	5.0400	167.4	0.0200	25.0	14.4	703.82	
553	5.0600	174.9	0.0200	25.0	7.5	456.60	
557	5.0800	179.5	0.0200	25.0	4.6	219.62	
562	5.1000	182.3	0.0200	25.0	2.8	85.71	
566	5.1200	184.5	0.0200	25.0	2.2		
569	5.1400	186.5	0.0200	25.0	2.0		
572	5.1600	188.3	0.0200	25.0	1.8		
576	5.1800	189.5	0.0200	25.0	1.2		
579	5.2000	190.8	0.0200	25.0	1.3		

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3

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Measured Values Scope 1/1, Sample 2/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-41.9		25.0			
12	2.0000	11.4	2.0000	25.0	53.3		
45	2.0200	10.7	0.0200	25.0	-0.7		
48	2.0400	10.9	0.0200	25.0	0.2		
52	2.0600	11.4	0.0200	25.0	0.5		
55	2.0800	11.4	0.0200	25.0	0.0	8.87	
58	2.1000	11.7	0.0200	25.0	0.3	15.44	
61	2.1200	12.0	0.0200	25.0	0.3	12.02	
64	2.1400	12.3	0.0200	25.0	0.3	11.45	
67	2.1600	12.5	0.0200	25.0	0.2	11.51	
70	2.1800	12.6	0.0200	25.0	0.1	10.21	
73	2.2000	12.9	0.0200	25.0	0.3	10.50	
76	2.2200	13.1	0.0200	25.0	0.2	10.91	
79	2.2400	13.3	0.0200	25.0	0.2	11.80	
82	2.2600	13.6	0.0200	25.0	0.3	12.08	
85	2.2800	13.8	0.0200	25.0	0.2	11.71	
88	2.3000	14.0	0.0200	25.0	0.2	11.80	
91	2.3200	14.3	0.0200	25.0	0.3	11.28	
94	2.3400	14.5	0.0200	25.0	0.2	10.13	
98	2.3600	14.7	0.0200	25.0	0.2	10.79	
101	2.3800	14.9	0.0200	25.0	0.2	10.73	
104	2.4000	15.1	0.0200	25.0	0.2	11.48	
107	2.4200	15.3	0.0200	25.0	0.2	11.80	
110	2.4400	15.7	0.0200	25.0	0.4	11.77	
113	2.4600	15.8	0.0200	25.0	0.1	11.80	
116	2.4800	16.1	0.0200	25.0	0.3	11.03	
119	2.5000	16.2	0.0200	25.0	0.1	9.67	
122	2.5200	16.5	0.0200	25.0	0.3	9.22	
125	2.5400	16.7	0.0200	25.0	0.2	10.00	
128	2.5600	16.8	0.0200	25.0	0.1	9.94	
131	2.5800	17.0	0.0200	25.0	0.2	10.57	

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134	2.6000	17.3	0.0200	25.0	0.3	10.19
137	2.6200	17.5	0.0200	25.0	0.2	11.29
140	2.6400	17.7	0.0200	25.0	0.2	11.32
143	2.6600	17.9	0.0200	25.0	0.2	10.19
146	2.6800	18.1	0.0200	25.0	0.2	9.99
149	2.7000	18.4	0.0200	25.0	0.3	10.51
152	2.7200	18.5	0.0200	25.0	0.1	10.42
156	2.7400	18.7	0.0200	25.0	0.2	10.79
159	2.7600	19.0	0.0200	25.0	0.3	11.57
162	2.7800	19.2	0.0200	25.0	0.2	11.72
165	2.8000	19.4	0.0200	25.0	0.2	11.29
168	2.8200	19.7	0.0200	25.0	0.3	10.28
171	2.8400	19.9	0.0200	25.0	0.2	9.64
174	2.8600	20.0	0.0200	25.0	0.1	10.22
177	2.8800	20.2	0.0200	25.0	0.2	11.08
180	2.9000	20.5	0.0200	25.0	0.3	11.20
183	2.9200	20.7	0.0200	25.0	0.2	11.78
186	2.9400	21.0	0.0200	25.0	0.3	11.89
189	2.9600	21.2	0.0200	25.0	0.2	10.42
192	2.9800	21.3	0.0200	25.0	0.1	9.70
195	3.0000	21.6	0.0200	25.0	0.3	9.50
198	3.0200	21.8	0.0200	25.0	0.2	9.90
201	3.0400	21.9	0.0200	25.0	0.1	11.03
204	3.0600	22.2	0.0200	25.0	0.3	11.57
207	3.0800	22.4	0.0200	25.0	0.2	10.90
210	3.1000	22.7	0.0200	25.0	0.3	12.08
214	3.1200	22.9	0.0200	25.0	0.2	12.02
217	3.1400	23.1	0.0200	25.0	0.2	11.70
220	3.1600	23.3	0.0200	25.0	0.2	12.08
223	3.1800	23.7	0.0200	25.0	0.4	11.99
226	3.2000	23.8	0.0200	25.0	0.1	11.71
229	3.2200	24.1	0.0200	25.0	0.3	11.01
232	3.2400	24.3	0.0200	25.0	0.2	10.98
235	3.2600	24.5	0.0200	25.0	0.2	11.44
238	3.2800	24.7	0.0200	25.0	0.2	11.99
241	3.3000	25.0	0.0200	25.0	0.3	12.56
244	3.3200	25.3	0.0200	25.0	0.3	12.44
247	3.3400	25.5	0.0200	25.0	0.2	13.27
250	3.3600	25.7	0.0200	25.0	0.2	13.62
253	3.3800	26.1	0.0200	25.0	0.4	12.89
256	3.4000	26.2	0.0200	25.0	0.1	13.27
259	3.4200	26.7	0.0200	25.0	0.5	13.04
262	3.4400	26.8	0.0200	25.0	0.1	13.18
265	3.4600	27.0	0.0200	25.0	0.2	13.55
268	3.4800	27.4	0.0200	25.0	0.4	14.34
272	3.5000	27.6	0.0200	25.0	0.2	13.12
275	3.5200	28.0	0.0200	25.0	0.4	13.58
278	3.5400	28.2	0.0200	25.0	0.2	13.00
281	3.5600	28.4	0.0200	25.0	0.2	12.37
284	3.5800	28.6	0.0200	25.0	0.2	13.26
287	3.6000	29.0	0.0200	25.0	0.4	13.71
290	3.6200	29.3	0.0200	25.0	0.3	16.08

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293	3.6400	29.5	0.0200	25.0	0.2	16.04
296	3.6600	29.9	0.0200	25.0	0.4	14.96
299	3.6800	30.2	0.0200	25.0	0.3	15.57
302	3.7000	30.6	0.0200	25.0	0.4	16.26
305	3.7200	30.7	0.0200	25.0	0.1	15.68
308	3.7400	31.1	0.0200	25.0	0.4	15.08
311	3.7600	31.6	0.0200	25.0	0.5	15.48
314	3.7800	31.7	0.0200	25.0	0.1	15.07
317	3.8000	32.0	0.0200	25.0	0.3	15.88
320	3.8200	32.4	0.0200	25.0	0.4	15.98
324	3.8400	32.7	0.0200	25.0	0.3	17.43
327	3.8600	33.0	0.0200	25.0	0.3	18.38
330	3.8800	33.5	0.0200	25.0	0.5	17.56
333	3.9000	33.8	0.0200	25.0	0.3	16.64
336	3.9200	34.1	0.0200	25.0	0.3	17.01
339	3.9400	34.4	0.0200	25.0	0.3	17.30
342	3.9600	34.8	0.0200	25.0	0.4	17.12
345	3.9800	35.1	0.0200	25.0	0.3	17.69
348	4.0000	35.6	0.0200	25.0	0.5	19.09
351	4.0200	35.9	0.0200	25.0	0.3	19.94
354	4.0400	36.2	0.0200	25.0	0.3	21.29
357	4.0600	36.7	0.0200	25.0	0.5	22.15
360	4.0800	37.3	0.0200	25.0	0.6	21.40
364	4.1000	37.6	0.0200	25.0	0.3	21.50
367	4.1200	38.1	0.0200	25.0	0.5	21.62
370	4.1400	38.4	0.0200	25.0	0.3	21.12
373	4.1600	38.8	0.0200	25.0	0.4	21.41
376	4.1800	39.4	0.0200	25.0	0.6	23.14
379	4.2000	39.9	0.0200	25.0	0.5	23.44
382	4.2200	40.2	0.0200	25.0	0.3	24.01
385	4.2400	40.7	0.0200	25.0	0.5	22.70
388	4.2600	41.3	0.0200	25.0	0.6	23.96
392	4.2800	41.7	0.0200	25.0	0.4	25.97
394	4.3000	42.2	0.0200	25.0	0.5	27.58
398	4.3200	42.7	0.0200	25.0	0.5	27.27
401	4.3400	43.5	0.0200	25.0	0.8	27.69
404	4.3600	43.9	0.0200	25.0	0.4	28.72
407	4.3800	44.5	0.0200	25.0	0.6	29.83
410	4.4000	45.0	0.0200	25.0	0.5	29.32
413	4.4200	45.7	0.0200	25.0	0.7	32.05
416	4.4400	46.4	0.0200	25.0	0.7	33.75
419	4.4600	47.1	0.0200	25.0	0.7	35.17
422	4.4800	47.6	0.0200	25.0	0.5	36.62
425	4.5000	48.7	0.0200	25.0	1.1	38.86
428	4.5200	49.1	0.0200	25.0	0.4	38.55
432	4.5400	50.2	0.0200	25.0	1.1	40.15
437	4.5600	50.9	0.0200	25.0	0.7	40.50
440	4.5800	51.8	0.0200	25.0	0.9	41.29
443	4.6000	52.3	0.0200	25.0	0.5	43.57
447	4.6200	53.6	0.0200	25.0	1.3	43.82
450	4.6400	54.3	0.0200	25.0	0.7	47.88
453	4.6600	55.4	0.0200	25.0	1.1	51.49

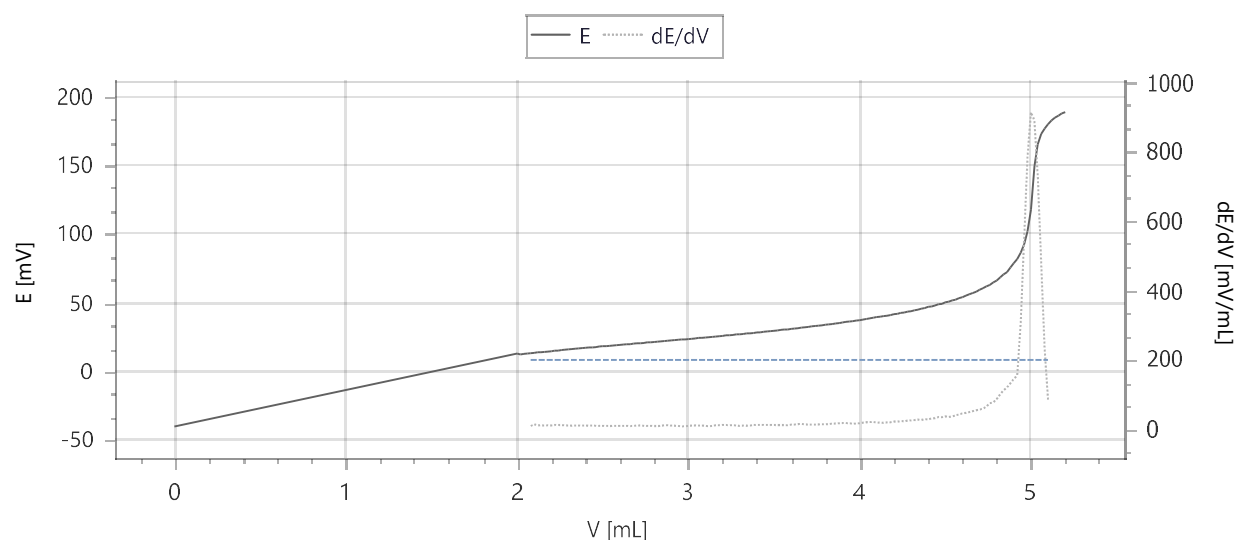
Titration report

LabX

456	4.6800	56.3	0.0200	25.0	0.9	54.96	
459	4.7000	57.5	0.0200	25.0	1.2	57.55	
462	4.7200	58.8	0.0200	25.0	1.3	62.90	
465	4.7400	60.1	0.0200	25.0	1.3	68.63	
468	4.7600	61.4	0.0200	25.0	1.3	75.47	
472	4.7800	63.0	0.0200	25.0	1.6	81.01	
476	4.8000	64.7	0.0200	25.0	1.7	88.01	
479	4.8200	66.8	0.0200	25.0	2.1	96.39	
483	4.8400	68.6	0.0200	25.0	1.8	105.55	
486	4.8600	70.8	0.0200	25.0	2.2	116.44	
491	4.8800	73.4	0.0200	25.0	2.6	132.28	
494	4.9000	76.4	0.0200	25.0	3.0	144.38	
498	4.9200	79.9	0.0200	25.0	3.5	142.61	
501	4.9400	84.3	0.0200	25.0	4.4	275.75	
508	4.9600	90.1	0.0200	25.0	5.8	517.87	
515	4.9800	98.0	0.0200	25.0	7.9	764.43	
528	5.0000	112.9	0.0200	25.0	14.9	923.44	
	5.012149	132.6				938.70	EQP
540	5.0200	145.3	0.0200	25.0	32.4	933.21	
545	5.0400	163.3	0.0200	25.0	18.0	791.55	
549	5.0600	172.2	0.0200	25.0	8.9	542.68	
553	5.0800	177.6	0.0200	25.0	5.4	275.14	
559	5.1000	180.5	0.0200	25.0	2.9	102.16	
562	5.1200	183.2	0.0200	25.0	2.7		
565	5.1400	185.2	0.0200	25.0	2.0		
568	5.1600	187.1	0.0200	25.0	1.9		
572	5.1800	188.6	0.0200	25.0	1.5		
574	5.2000	189.8	0.0200	25.0	1.2		

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3



Measured Values Scope 1/1, Sample 3/3

Time	V	Meas. value	dV	Temp.	dE	dE/dV	EQP
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Titration report

LabX

[s]	[mL]	[mV]	[mL]	[°C]	[mV]	[mV/mL]
0	0.0000	-39.8		25.0		
11	2.0000	13.1	2.0000	25.0	52.9	
44	2.0200	12.5	0.0200	25.0	-0.6	
48	2.0400	12.9	0.0200	25.0	0.4	
51	2.0600	13.1	0.0200	25.0	0.2	
54	2.0800	13.5	0.0200	25.0	0.4	10.26
57	2.1000	13.6	0.0200	25.0	0.1	15.31
60	2.1200	14.0	0.0200	25.0	0.4	11.72
63	2.1400	14.2	0.0200	25.0	0.2	12.27
66	2.1600	14.4	0.0200	25.0	0.2	11.38
69	2.1800	14.6	0.0200	25.0	0.2	12.04
72	2.2000	14.9	0.0200	25.0	0.3	11.72
75	2.2200	15.2	0.0200	25.0	0.3	13.28
78	2.2400	15.3	0.0200	25.0	0.1	13.02
81	2.2600	15.7	0.0200	25.0	0.4	12.56
84	2.2800	15.9	0.0200	25.0	0.2	11.99
87	2.3000	16.2	0.0200	25.0	0.3	11.44
90	2.3200	16.3	0.0200	25.0	0.1	10.98
93	2.3400	16.6	0.0200	25.0	0.3	11.30
96	2.3600	16.8	0.0200	25.0	0.2	11.71
99	2.3800	17.0	0.0200	25.0	0.2	11.19
102	2.4000	17.3	0.0200	25.0	0.3	11.37
106	2.4200	17.6	0.0200	25.0	0.3	10.93
108	2.4400	17.6	0.0200	25.0	0.0	11.51
112	2.4600	17.9	0.0200	25.0	0.3	10.79
115	2.4800	18.2	0.0200	25.0	0.3	10.45
118	2.5000	18.4	0.0200	25.0	0.2	10.74
121	2.5200	18.6	0.0200	25.0	0.2	10.51
124	2.5400	18.7	0.0200	25.0	0.1	9.99
127	2.5600	19.0	0.0200	25.0	0.3	10.51
130	2.5800	19.2	0.0200	25.0	0.2	10.69
133	2.6000	19.4	0.0200	25.0	0.2	10.80
136	2.6200	19.7	0.0200	25.0	0.3	10.80
139	2.6400	19.8	0.0200	25.0	0.1	10.71
142	2.6600	20.0	0.0200	25.0	0.2	9.93
145	2.6800	20.3	0.0200	25.0	0.3	10.53
148	2.7000	20.5	0.0200	25.0	0.2	11.54
151	2.7200	20.7	0.0200	25.0	0.2	12.01
154	2.7400	20.8	0.0200	25.0	0.1	10.77
157	2.7600	21.3	0.0200	25.0	0.5	10.28
160	2.7800	21.4	0.0200	25.0	0.1	10.42
163	2.8000	21.5	0.0200	25.0	0.1	9.96
166	2.8200	21.7	0.0200	25.0	0.2	10.04
170	2.8400	22.0	0.0200	25.0	0.3	10.42
173	2.8600	22.2	0.0200	25.0	0.2	12.30
176	2.8800	22.4	0.0200	25.0	0.2	12.09
179	2.9000	22.7	0.0200	25.0	0.3	11.28
182	2.9200	22.9	0.0200	25.0	0.2	11.00
185	2.9400	23.1	0.0200	25.0	0.2	10.23
188	2.9600	23.3	0.0200	25.0	0.2	8.89
191	2.9800	23.5	0.0200	25.0	0.2	9.50

Titration report

LabX

194	3.0000	23.7	0.0200	25.0	0.2	10.74
197	3.0200	23.8	0.0200	25.0	0.1	11.49
200	3.0400	24.1	0.0200	25.0	0.3	11.86
203	3.0600	24.5	0.0200	25.0	0.4	11.48
206	3.0800	24.6	0.0200	25.0	0.1	11.22
209	3.1000	24.8	0.0200	25.0	0.2	10.82
212	3.1200	25.0	0.0200	25.0	0.2	9.60
215	3.1400	25.2	0.0200	25.0	0.2	10.70
218	3.1600	25.5	0.0200	25.0	0.3	12.60
221	3.1800	25.7	0.0200	25.0	0.2	12.71
224	3.2000	25.9	0.0200	25.0	0.2	13.08
228	3.2200	26.3	0.0200	25.0	0.4	13.08
231	3.2400	26.5	0.0200	25.0	0.2	12.42
234	3.2600	26.7	0.0200	25.0	0.2	12.59
237	3.2800	27.0	0.0200	25.0	0.3	11.50
240	3.3000	27.2	0.0200	25.0	0.2	10.61
243	3.3200	27.4	0.0200	25.0	0.2	11.30
246	3.3400	27.7	0.0200	25.0	0.3	10.92
249	3.3600	27.8	0.0200	25.0	0.1	11.70
252	3.3800	28.1	0.0200	25.0	0.3	13.11
255	3.4000	28.4	0.0200	25.0	0.3	13.48
258	3.4200	28.6	0.0200	25.0	0.2	13.50
262	3.4400	29.0	0.0200	25.0	0.4	13.58
265	3.4600	29.2	0.0200	25.0	0.2	13.21
268	3.4800	29.4	0.0200	25.0	0.2	13.53
271	3.5000	29.7	0.0200	25.0	0.3	13.27
274	3.5200	30.0	0.0200	25.0	0.3	12.69
277	3.5400	30.3	0.0200	25.0	0.3	13.01
280	3.5600	30.5	0.0200	25.0	0.2	12.72
283	3.5800	30.7	0.0200	25.0	0.2	12.42
286	3.6000	31.0	0.0200	25.0	0.3	14.07
289	3.6200	31.3	0.0200	25.0	0.3	14.43
292	3.6400	31.6	0.0200	25.0	0.3	15.55
295	3.6600	31.9	0.0200	25.0	0.3	15.90
298	3.6800	32.3	0.0200	25.0	0.4	14.92
302	3.7000	32.4	0.0200	25.0	0.1	14.48
305	3.7200	32.9	0.0200	25.0	0.5	14.31
308	3.7400	33.1	0.0200	25.0	0.2	14.40
311	3.7600	33.3	0.0200	25.0	0.2	14.74
314	3.7800	33.7	0.0200	25.0	0.4	15.06
317	3.8000	34.0	0.0200	25.0	0.3	14.61
320	3.8200	34.3	0.0200	25.0	0.3	16.02
324	3.8400	34.6	0.0200	25.0	0.3	16.79
326	3.8600	34.9	0.0200	25.0	0.3	17.51
330	3.8800	35.3	0.0200	25.0	0.4	17.99
333	3.9000	35.7	0.0200	25.0	0.4	18.85
336	3.9200	36.1	0.0200	25.0	0.4	18.53
339	3.9400	36.4	0.0200	25.0	0.3	17.70
342	3.9600	36.7	0.0200	25.0	0.3	16.71
345	3.9800	37.2	0.0200	25.0	0.5	17.20
348	4.0000	37.4	0.0200	25.0	0.2	18.14
351	4.0200	37.8	0.0200	25.0	0.4	20.09

Titration report

LabX

354	4.0400	38.2	0.0200	25.0	0.4	21.07	
358	4.0600	38.7	0.0200	25.0	0.5	21.50	
361	4.0800	39.1	0.0200	25.0	0.4	21.87	
364	4.1000	39.6	0.0200	25.0	0.5	19.93	
367	4.1200	39.9	0.0200	25.0	0.3	20.28	
370	4.1400	40.3	0.0200	25.0	0.4	19.38	
373	4.1600	40.7	0.0200	25.0	0.4	20.73	
376	4.1800	41.1	0.0200	25.0	0.4	20.95	
380	4.2000	41.7	0.0200	25.0	0.6	23.05	
383	4.2200	41.9	0.0200	25.0	0.2	22.95	
386	4.2400	42.6	0.0200	25.0	0.7	24.07	
389	4.2600	42.9	0.0200	25.0	0.3	23.91	
393	4.2800	43.6	0.0200	25.0	0.7	25.80	
396	4.3000	43.9	0.0200	25.0	0.3	26.22	
399	4.3200	44.6	0.0200	25.0	0.7	27.27	
402	4.3400	45.0	0.0200	25.0	0.4	28.60	
406	4.3600	45.8	0.0200	25.0	0.8	27.54	
409	4.3800	46.1	0.0200	25.0	0.3	29.92	
413	4.4000	47.0	0.0200	25.0	0.9	28.83	
416	4.4200	47.3	0.0200	25.0	0.3	31.99	
419	4.4400	48.0	0.0200	25.0	0.7	32.52	
422	4.4600	48.9	0.0200	25.0	0.9	35.85	
425	4.4800	49.3	0.0200	25.0	0.4	35.11	
429	4.5000	50.4	0.0200	25.0	1.1	37.72	
432	4.5200	50.8	0.0200	25.0	0.4	36.05	
435	4.5400	51.8	0.0200	25.0	1.0	37.35	
438	4.5600	52.3	0.0200	25.0	0.5	40.55	
442	4.5800	53.4	0.0200	25.0	1.1	43.47	
445	4.6000	54.0	0.0200	25.0	0.6	46.47	
449	4.6200	55.2	0.0200	25.0	1.2	47.78	
454	4.6400	56.2	0.0200	25.0	1.0	49.99	
458	4.6600	57.2	0.0200	25.0	1.0	52.66	
461	4.6800	58.0	0.0200	25.0	0.8	55.24	
465	4.7000	59.4	0.0200	25.0	1.4	57.35	
468	4.7200	60.5	0.0200	25.0	1.1	60.17	
472	4.7400	62.0	0.0200	25.0	1.5	66.04	
476	4.7600	63.0	0.0200	25.0	1.0	75.71	
479	4.7800	64.7	0.0200	25.0	1.7	78.03	
482	4.8000	66.2	0.0200	25.0	1.5	87.69	
485	4.8200	68.6	0.0200	25.0	2.4	100.85	
490	4.8400	70.6	0.0200	25.0	2.0	112.40	
493	4.8600	72.1	0.0200	25.0	1.5	121.10	
497	4.8800	75.8	0.0200	25.0	3.7	133.24	
502	4.9000	78.8	0.0200	25.0	3.0	145.82	
505	4.9200	81.8	0.0200	25.0	3.0	155.87	
510	4.9400	86.4	0.0200	25.0	4.6	305.35	
513	4.9600	92.6	0.0200	25.0	6.2	546.07	
518	4.9800	101.8	0.0200	25.0	9.2	784.75	
528	5.0000	117.6	0.0200	25.0	15.8	913.17	
	5.007243	129.2				915.42	EQP
538	5.0200	149.5	0.0200	25.0	31.9	889.45	
542	5.0400	165.0	0.0200	25.0	15.5	727.84	

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546	5.0600	172.7	0.0200	25.0	7.7	479.14
553	5.0800	176.5	0.0200	25.0	3.8	231.07
556	5.1000	179.7	0.0200	25.0	3.2	84.84
560	5.1200	182.4	0.0200	25.0	2.7	
563	5.1400	184.4	0.0200	25.0	2.0	
566	5.1600	186.0	0.0200	25.0	1.6	
569	5.1800	187.5	0.0200	25.0	1.5	
572	5.2000	188.4	0.0200	25.0	0.9	

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1
Sample ID 2
Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
04.02.2026 08:54:00	PredefinedUser	Administrator